

433-HW7

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Problem 1

```
# 3 calls expected over 1 minute, 15 calls expected over 5 minutes  
# X~Poi(15)  
# P(X=8)  
dpois(8,15)
```

```
## [1] 0.0194443
```

Problem 2

```
bp.app = function(x,n,p) {  
  lambda = n*p # variable for lambda  
  b_total_prob = choose(n,x)*p^x*(1-p)^(n-x) # Binomial P(X=x)  
  p_total_prob = exp(-lambda)*lambda^x/factorial(x) # Poisson P(X=x)  
  for (i in 0:(x-1)) {  
    # discrete CDFs are sums, use loop to calculate and add chance of each x  
    b_total_prob = b_total_prob + choose(n,i)*p^i*(1-p)^(n-i)  
    p_total_prob = p_total_prob + exp(-lambda)*lambda^i/factorial(i)  
  }  
  return(c(b_total_prob,p_total_prob))  
}
```

```
bp.app(20,100,0.2) # both should be around ~0.559
```

```
## [1] 0.5594616 0.5590926
```

Problem 3

```
binom.pmf = function(x, k, p, multiple=1) {  
  # x = num successes, k = num trials, p = chance of success,  
  # multiple = default value, updated through recursion
```

```

if (x == 0) { # base case
  return(multiple*(1-p)^k) # f(0,k,p)
}
else { # recursive part to determine probability
  # update multiple value with new x, times current multiple (Bin combos)
  multiple = multiple*(k-x+1)*p/(x*(1-p))
  return(binom.pmf(x-1, k, p, multiple)) # call function again with f(x-1,k,p)
}
}

```

```

binom.pmf(3,12,0.5) # 3 successes in 12 trials, chance of success 1/2

```

```

## [1] 0.05371094

```

```

dbinom(3,12,0.5) # should be ~0.053

```

```

## [1] 0.05371094

```

Problem 4

```

# X~U(0,1), Y = X^2
# Use cdf of X to show P(Y <= y) = sqrt(y) for 0 < y < 1, obtain pdf of Y
# Evaluate E(Y) and Var(Y)

# cdf x: 0 for x < 0
#         x/1 for 0 <= x <= 1
#         1 for x > 1
# P(Y <= y) = P(X^2 <= y) = P(-sqrt(y) < x < sqrt(y))
# plug into cdf: (sqrt(y) - (-sqrt(y))) / (1 - (-1)) = sqrt(y)
# differentiate cdf of Y to get pdf of Y:
# pdf y: 1/(2sqrt(y)) for 0 <= y <= 1
#         0 otherwise
# X only exists on interval from 0 to 1, any value within this interval squared
# is also between 0 and 1, thus Y has the same range of values
# E(Y) = integral of y*y^(-1/2)/2 from 0 to 1
#       = y^(3/2)/3 evaluated from 0 to 1 = 1/3 - 0/3 = 1/3
# Var(Y) = E(Y^2)-E(Y)^2
#         = integral of y^2*y^(-1/2)/2 from 0 to 1 minus (1/3)^2
#         = 1/5 - 1/9 = 4/45

```